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1BG235 17429: Evolution and diversity of microbial eukaryotes

Batrachochytrium dendrobatidis

Batrachochytrium dendrobatidis (Bd, figure 1) is a zoospore creating fungi part of the phylum Chytridiomycota, under the subphylum Chytridiomycetes (Sewell et al. 2021). As most Chytrids it's an aquatic fungi and it's the pathogen behind chytridiomycosis, which is a skin disease that affects amphibians (Sewell et al. 2021). The life cycle (figure 2) of Bd consists of two stages, the first is a zoosporic stage where the zoospore encyst and absorbs the flagella to create rhizoids (figure 1). The rhizoids penetrate the skin of an amphibian and matures into a thallus which in turn matures into a sporangium from where new zoospores are created (Voyles et al. 2011).

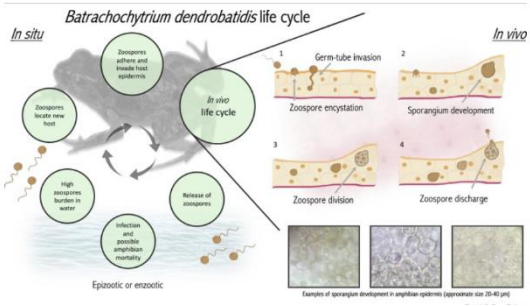


Figure 2: The life cycle of Batrachochytrium denrobatidis. It shows how it consists of two stages, a In situ zoosporic stage and an In vivo germination stage.

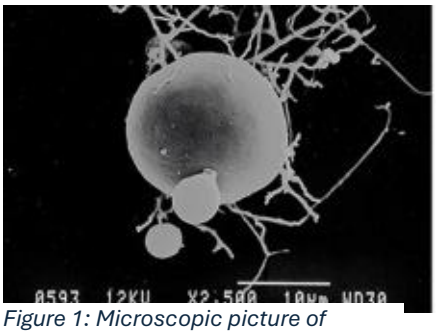


Figure 1: Microscopic picture of Batrachochytrium dendrobatidis, the picture shows a thallus with rhizoids.

The first *Bd* was discovered on a blue poison dart frog (*Dendrobrates azureus)*, however today multiple genetic lineages have been discovered (figure 3) from different amphibians. The spread is believed to be caused by *BD*GPL (*Batrachochytrium dendrobatidits* Global Panzootic Lineage), a genotype with a high spreading ability (Sewell et al.2011). Six main lineages have been discovered, four are found in Asia while the rest of the world tend to have two lineages (figure 3), indicating that the original *BD* strain have originated from Asia (Sewell et al. 2011).

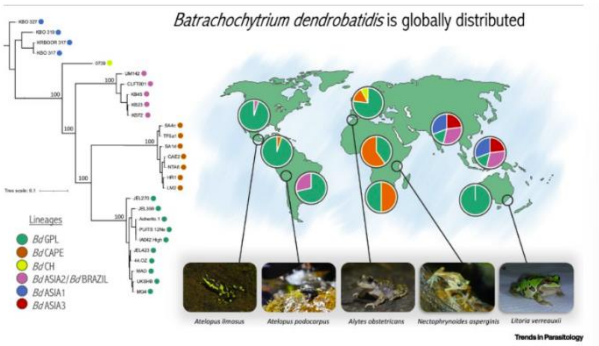


Figure 3: A figure which show the six lineages of Batrachochytrium dendrobatidis (the green colour represents the Global panzootic lineage, pink, blue and red is different Asian lineage, the orange is African lineage, and the yellow is the Swiss/European). It also shows a map where each lineage can be found.

The disease is a global problem, and it has led to some species extinctions. It works by infecting amphibians with their zoospores. When they penetrate the skin, the sporangium matures inside the skin and blocks the pores (figure 2) which the amphibians use for osmoregulation (Sewel et al. 2011, Larsen et al. 2021). This creates an electrolyte disruption and there are recorded cases where asystolic cardiac arrest happened (Sewel et. Al 2011). The cardiac arrest is caused by reductions of electrolytes in the blood (plasma) and in turn the electrical pulse in the heart worsens until it stops completely (Voyles et al. 2011).

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References:

Thomas R. Sewell, Joyce Longcore, Matthew C. Fisher, 2021, *Batrachochytrium dendrobatidis, Trends in parasitology, volume 37, Issue 10, ELSEVIER: ScienceDirect* https://doi.org/10.1016/j.pt.2021.04.014

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Picture sources:

Thomas R. Sewell, Joyce Longcore, Matthew C. Fisher, 2021, *Batrachochytrium dendrobatidis, Trends in parasitology, volume 37, Issue 10, ELSEVIER: ScienceDirect* https://doi.org/10.1016/j.pt.2021.04.014